



STANDARD COLLEGE NTUNGAMO
SENIOR FOUR INTERNAL MOCK EXAMINATIONS

Uganda Certificate of Education

PHYSICS

Theory

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

- Attempt **five** items
- Attempt **all** items in section **A** and **two** in section **B**. In section B attempt **one** in each part.
- Begin each item on a fresh page

SECTION A

This section consists of three COMPULSORY items.

ITEM 1

Your area Member of Parliament has organized a camp in a nearby playing ground that is located between a tall building and a cliff to give feed back to his people. He is also required to donate some equipment's to bodaboda riders who drive during the day and night in a bid to reduce accidents they normally meets on the road while on their motorcycles. Among the items to select from by bodaboda riders were concave mirrors, convex lenses, small plane mirrors.

During the presentations, different reactions and observations were made and noticed by the Member of Parliament.

- i) One herds man threw a grazing stick to a cow that accidentally fell in a nearby a pond with clear waters. As he proceeded to pick it he found out that a big part of it appeared as if it's broken and his appearance appeared in water. This confused him.
- ii). One boda boda rider required for some mirrors to use while riding and the front lighting system as the ones originally on the motorcycle were spoilt when it fell down.
- iii) Some people in attendance are worried that the sound produced by the loud speakers could pose risks to their ears because its too loud, unclear and disorganized.

Fig 1 (a) shows the rarefactions and compressions produced by the sound wave heard

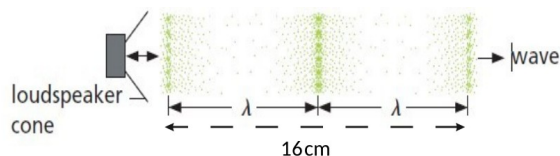
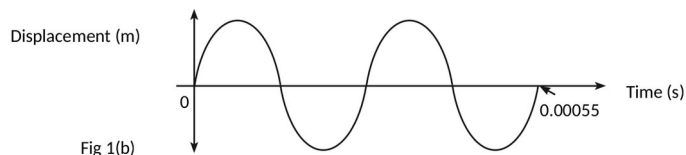


Figure 1 (b) shows the wave signal that produces the sound



Audible frequency range = 20 - 20000 Hz

Use speed of sound in air = 330 ms^{-1}

Task

- To arrest the herds man concerns/ fears
- Select with reason, the most appropriate mirror equipment to donate to the boda boda rider.
- Understand whether the sound produced posed risks to the attendants' ears.
- why the sound seemed unclear and disorganized.
- Describe one area in which one of the concepts can be applied in daily life.

ITEM 2:

Recently, your female friend enrolled in an Astrophysics course in one of the outside countries. After a few days of settling in, she called her parents in Uganda by 9 am local time in that country. However, in Uganda, it was 8 pm. Her parents had just received the news that the full moon had been sighted using satellite communication, indicating the start of fasting for the holy month. During the phone conversation, she mentioned that her first lecture was about the life cycle of a star. This interested the parents who wondered how a star could have a life cycle and they were also curious about the significant time difference between the two countries.

Task:

As a physics learner, help the parents to understand;

- What causes the differences in time between the two countries?
- The different phases of the moon.
- The life cycle of a star.
- How it is possible to talk to her parents.

ITEM 3:

In a certain town, the amount of water reaching people for use in homes has reduced and they are worried of water scarcity in the near future if this is not rectified. The authorities suspect a leakage in underground pipe and would like to locate the pipe spot.

They have been told that radioactive material should be used to check the spot but they have little knowledge of doing it.

Table 1 shows the materials available for use in the process of checking if there is leakage underground.

Detector	Counter
Spark counter, cloud chamber , geiger muller tube	Rate meter, scaler.

A sample of Technetium-99 of activity 200 counts per minute is used in the detector until its activity reduces to background radiation of level of 50 counts per minute.

The locals have been asked to wait for sometime when this material is used and also to take precautionary measures of dangers associated with the use of the material but are left wondering when.

Hint

Half life of Technetium is 6 hours

Task.

You have been approached by both locals and local authorities, help them with relevant information regarding to the challenges faced.

SECTION B: PART 1

Answer **one item** from this part

Item 4:

In a certain village, the roads are made of marram, with many potholes. On a certain rainy day, a driver of a car which has a mass of 1000kg traveling at 60kmh^{-1} sets off on a journey but along the way, the driver notices a big object on the road and suddenly brings the car to rest. In the process, the car skidded off the road and got stuck.

Whenever the braking force is applied in such a manner, about 60% of the kinetic energy is expended in the brake drum. The material of the drum and brake pads can absorb heat up to 150 kJ of heat energy, beyond which the tyres will burn.

Task:

As a learner of physics, help the driver;

- a. Understand how to avoid similar occurrence.
- b. To understand why the car tyres did not burn.

Item 5:

A tourist who weighs 70kg, before climbing a mountain is availed with two sets of shoes; one with sharp spikes and the other with flat soles.

While climbing the mountain, the tourist follows a slippery road and had difficulty in climbing.

Along the way, the tourist records a barometric reading at location A and B as 75cmHg and 71.7cmHg and wonders how far apart they are. The tourist felt very exhausted when arriving at location B and at the same time, he had nose bleeding. This scared him more. Later he also had difficulty in having the food ready.

Hint;

Density of air 1.25kgm^{-3}

Density of mercury = $1.36 \times 10^4 \text{ kgm}^{-3}$

Task:

As a learner of physics, help the tourist understand why;

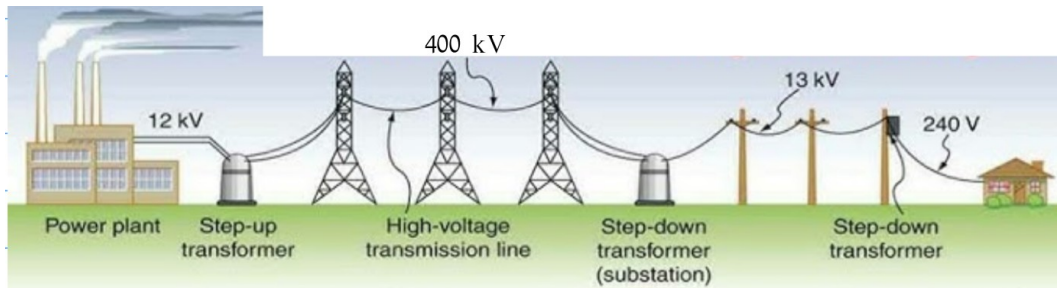
- a. There was difficulty in climbing the mountain and how far apart the two points were.
- b. He was very tired and the cause of nose bleeding.
- c. There was difficulty in getting the food ready.

PART 2

Attempt any **one item** from this section.

Item 6:

In a certain village, electricity at a substation is transmitted at 13 kV with a current of 0.05 A using thick aluminum wires for use inside a house at 240 V.



One of the villagers wanted to install a television set (TV) in his house. He approached a technician who told him that a TV operates on direct current of 4A and 20% of the electrical energy is lost during voltage change. The villager did not understand what he meant and wondered if the TV will still work.

Task

Use your knowledge of Physics to help the villager;

- Understand how the voltage is changed from 13 kV to 240 V.
- Why thick aluminum wires are used during power transmission?
- Determine if television sets will still work.
- Explain how direct current (DC) was obtained.

Item 7:

A group of students in a certain boarding school decided to revise for exams at night from a field north of the school. They were provided with a voltmeter, three DC electric lamps of resistance 12Ω each, an accumulator rating 12v and 8A, a switch, connecting wires, 3-inch steel nail, a bare copper wire of length 50cm and a piece of thread. However, the rest of your colleagues are unable to connect the circuit correctly except you. They also want to know the best ways of using the accumulator for a longer time. Further after revision, they could not easily tell the direction from which they came.

Task.

- Advise on how to generate light so that you and your colleagues can revise for a longer time and state assumptions made.
- Come up with advice that would enable you and your colleagues back to school using available materials.
- State the applications of the concepts observed in the scenario.

THE END

